Chemical Equilibrium Lectures

Prof. Dr. Ahmed Attia Abdelgaber

المحاضرة (٢)

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Equilibrium Laws for Gaseous Reactions

When all the **reactants and products are gases**,

We can formulate **mass action expressions in terms of partial pressures** as well as **molar concentrations**.

molar concentration of a **gas is proportional** to its **partial pressure**.

This comes from the ideal gas law,

$$PV = nRT$$

Solving for P gives:

$$P = \left(\frac{n}{V}\right)RT$$

The quantity (**n/V**) **has units of mol/L** and is the **molar concentration**.

P = (molar concentration) x RT

Write the **mass action expression** for reactions between gases in terms either of **molarities** **or, of** **partial pressures**.

When **molar concentrations** are used, we use the symbol K_c

When **partial pressures** are used, then K_p is the symbol.

For example,

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

Can be written in either of the following two ways:

$$\frac{[NH_3]^2}{[N_2][H_2]^3} = K_c \quad \left(\begin{array}{c}\text{because molar concentrations are used}\\ \text{in the mass action expression}\end{array}\right)$$

$$\frac{P^2_{NH_3}}{P_{N_2}P^3_{H_2}} = K_p \quad \left(\begin{array}{c}\text{because partial pressures are used in}\\ \text{the mass action expression}\end{array}\right)$$

The equilibrium molar concentrations can be used to calculate K_c, where the equilibrium partial pressures can be used to calculate K_p.

Ex.

Write the expression for K_p for the reaction

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

SOLUTION For K_p we use partial pressures in the mass action expression.

$$K_p = \frac{P^2_{NO_2}}{P_{N_2O_4}}$$

The Significance of the Magnitude of K

the size of the equilibrium constant gives us a measure of how far the reaction proceed toward completion when equilibrium is reached.

$$2H_2(g) + O_2(g) = 2H_2O(g)$$

has **$K_c = 9.1 \times 10^{80}$** at 25 °C.

$$K_c = \frac{[H_2O]^2}{[H_2]^2[O_2]} = \frac{9.1 \times 10^{80}}{1}$$

numerator to be **so much larger than the denominator**

concentration of H_2O to be enormous **in comparison** to the **concentrations of H_2 and O_2.**

at **equilibrium** **most of the** **hydrogen** **and** **oxygen** **atoms** **in the system** **are found in the H_2O** **and** **very few** **are present in** **H_2 and O_2** .

Thus, **the large value of K_c tells us** that **the reaction** between **H_2 and O_2** goes essentially to **completion**.

The reaction between N_2 and O_2 to give NO

$$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$$

has a very small equilibrium;

$$K_c = 4.8 \times 10^{-31} \text{ at } 25\ ^\circ\text{C}$$

The equilibrium law for this reaction is:

$$\frac{[NO]^2}{[N_2][O_2]} = 4.8 \times 10^{-31}$$

Since $10^{-31} = 1/10^{31}$, we can write this as

$$\frac{[NO]^2}{[N_2][O_2]} = \frac{4.8}{10^{31}}$$

the denominator is huge compared with the numerator,

very little NO is formed.

The reaction hardly proceeds at all **toward completion** before equilibrium is reached.

When K is very large

The reaction proceeds far toward completion. The position of equilibrium lies far toward the products.

When K ≈1

The concentrations of reactants and products are nearly the same at equilibrium. The position of equilibrium lies midway between reactants and products.

When K is very small

Extremely small amounts of products are formed. The position of equilibrium lies far toward the reactants.

The Relationship between K_p and K_c

P = (molar concentration) x RT

$$K_p = K_c(RT)^{\Delta n_g}$$

Δn_g is equal to the change in the number of moles of gas in going from the reactants to the products.

Δn_g = (moles of gaseous products) - (moles of gaseous reactants)

We use the coefficients of the balanced equation for the reaction to calculate the numerical value of Δn_g.

For example

$$N_2\,(g) + 3H_2(g) = 2NH_3(g)$$

Δn_g for this reaction equals -2.

$$2\,HI(g) = H_2(g) + I_2(g)$$

$\Delta n_g = 0$. Since (RT) raised to the zero power is equal to 1, $K_p = K_c$

EX

At 500 °C, the reaction between N_2 and H_2 to form ammonia has $K_c = 6.0 \times 10^{-2}$.

$$N_2(g) + 3H_2(g) = 2NH_3(g)$$

What is the numerical value of K_p for this reaction?

The equation that we wish to use is :

$$K_p = K_c(RT)^{\Delta n_g}$$

$\Delta n_g = -2$

R = 0.0821 L atm. mol $^{-1}$ K^{-1},

$K_c = 6.0 \times 10^{-2}$ $\Delta n_g = -2$

T = (500+273)K=773 K R= 0.082L atm mol^{-1} K^{-1}

Substituting these into the equation for K_p gives

$$K_p = (6.0 \times 10^{-2}) \times [(0.0821) \times (773)]^{-2}$$
$$= (6.0 \times 10^{-2}) \times (63.5)^{-2}$$
$$= 1.5 \times 10^{-5}$$

<u>In this case, K_p has a numerical value quite different from that of K_c</u>.

EX.

At 25°C, K_p for the reaction: **$N_2O_4(g) = 2NO_2(g)$** has a value of 0.140. Calculate the value of K_c.

$$K_p = K_c(RT)^{\Delta n_g}$$

This time, $\Delta n_g = 2 - 1 = +1$. Now let's tabulate the data.

K_p=0.140	Δn_g =+1
T=298K	R=0.082l L atm $mol^{-1}K^{-1}$

Solving the equation for K_c gives

$$K_c = \frac{K_p}{(RT)^{\Delta n_g}}$$

Substituting values into this equation yields

$$K_c = \frac{0.140}{[(0.0821) \times (298)]^1}$$
$$= 5.72 \times 10^{-3}$$

Once again, there is a substantial difference between the values of K_p and K_c.

Heterogeneous Equilibria

In a **homogeneous reaction** - or a **homogeneous equilibrium, all of the reactants and products are in the same phase**.

When more than one phase exists in a reaction mixture, we call it a heterogeneous reaction.

$$2NaHCO_3(s) \longrightarrow Na_2CO_3(s) + H_2O(g) + CO_2(g)$$

Heterogeneous reactions are able to reach equilibrium, just **as homogneous** reactions can.

$$2NaHCO_3(s) \rightleftharpoons Na_2CO_3(s) + H_2O(g) + CO_2(g)$$

1 mol $NaHCO_3$

38.9 cm^3

$$\text{Molarity} = \frac{1 \text{ mol } NaHCO_3}{0.0389 \text{ L}}$$

= 25.7 mol/L

2 mol $NaHCO_3$

77.8 cm^3

$$\text{Molarity} = \frac{2 \text{ mol } NaHCO_3}{0.0778 \text{ L}}$$

= 25.7 mol/L

$$\frac{[Na_2CO_3(s)]\,[H_2O(g)]\,[CO_2(g)]}{[NaHCO_3(s)]^2} = K$$

the concentration of a pure liquid or solid is unchangeable;

that is, **for any pure liquid or solid, the ratio of amount of substance to volume of substance is a constant**.

$$\frac{1 \text{ mol}}{0.0389 \text{ L}} = \frac{2 \text{ mol}}{0.0778 \text{ L}} = 25.7 \text{ mol/L}$$

$$[H_2O(g)][CO_2(g)] = \frac{K[NaHCO_3(s)]^2}{[Na_2CO_3(s)]} = K_c$$

The equilibrium law for a heterogeneous reaction is written without concentration terms for pure solids or liquids.

EX

The air pollutant sulfur dioxide can be removed from a gas mixture by passing mixture over calcium oxide. The equation is:

$$\mathbf{CaO(s) + SO_2(g) = CaSO_3(s)}$$

Write the equilibrium law for this system.

$$\frac{1}{[SO_2(g)]} = K_c$$

Le Châtelier's Principle and Chemical Equilibria

If we were pring ammonia, NH_3, by the reaction of N_2 with H_2, we might want to know to maximize the yield of NH_3.

Le Chatelier's principle:

allows us to predict the effects of outside influences on equilibria

Le Châtelier's Principle
States that:

If an outside influence upsets an equilibrium, the system undergoes a change in a direction counteracts the disturbing influence, and returns the system to equilibrium.

Outside influences:

1. Adding or removing a reactant or product.

If we change one of the concentrations in the system.

The result is that the **equilibrium is upset**.

For the system to return to equilibrium, the concentrations must change in a way that causes Q to equal K_c again.

As an example, let's study the equilibrium between two ions of copper.

$$Cu(H_2O)_4^{+2}(aq) + 4C1^-(aq) = CuCl_4^{-2}(aq) + 4H_2O$$

blue **yellow**

As noted, $Cu(H_2O)_4^{+2}$ is blue and $CuCl_4^{-2}$ is yellow. Mixtures of the two have an intermediate color and therefore appear blue green.

after the addition of water. It is blue because the equilibrium has shifted toward $Cu(H_2O)_4^{+2}$.

If we add $C1^-$ to an equilibrium mixture of these copper ions, the system.**This gives more $CuCl_4^{-2}$**, and the **equilibrium has shifted to the right**.

2- Changing the volume in gaseous reactions.

$$3H_2(g) + N_2(g) = 2NH_3(g)$$

If we reduce the volume of the reaction mixture, we expect the pressure to increase.

this equilibrium is shifted to the right when the volume of the reaction mixture is reduced.

Now let's look at the equilibrium

$$H_2(g) + I_2(g) = 2HI(g)$$

there is no change in the number of molecules of gas.This reaction, then, cannot respond to pressure changes,

changing the volume of the reaction vessel has no effect on the equilibrium.

3. Changes of temperature.

$$3H_2(g) + N_2(g) = 2NH_3(g) + heat$$

raising the temperature shifts this equilibrium to the left.

In general,
Increasing the temperature shifts a reaction in a direction that produces an endothermic change.

When the temperature of an exothermic reaction is increased, the value of the equilibrium constant becomes smaller.

Of course, just the opposite occurs for an endothermic reaction.

4. Effect of a catalyst.

catalysts do not affect the position of equilibrium in a system. The reason is that a catalyst affects both the forward and reverse reactions equally.

Both are speeded up to increase the same degree,

so adding a catalyst to a system has no net effect on the system's equilibrium composition.

The catalyst's only effect is to bring the system to equilibrium faster.

5. Addition of an inert gas at constant volume.

If this gas cannot react with the gases already present (i.e.,if the added gas is inert toward the substances in equilibrium), **the concentration of the reactants and products can't change**.

The concentrations will continue to satisfy the equilibrium law and

the reaction quotient will continue to equal K_c

so there will be **no change in the position of equilibrium**.

EX: $N_2O_4(g) = 2NO_2(g)$ is endothermic, with $\Delta H° =+ 56.9$ kJ. How will the amount of NO_2 at equilibtium be affected by:
(a) adding N_2O_4,
(b) lowering the pressure by increasing the volume of the container,
(c) raising the temperature, and
(d) adding a catalyst to the system? Which of these changes will after the value of K_c